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CHAPTER 5-Logu

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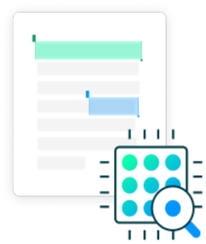

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CHAPTER 5

Results and Discussions

5.1 Introduction

This chapter presents a comprehensive experimental evaluation of the proposed blockchain-enabled privacy-preserving authentication frameworks developed in this research. The experimental setup, including dataset configuration, simulation environment, parameter settings, and implementation details, is first described to ensure reproducibility and fairness in evaluation. Subsequently, the performance of the proposed models is analyzed using standard metrics such as accuracy, sensitivity, specificity, F1-score, ROC characteristics, computational time, memory usage, encryption/decryption cost, and security robustness. A detailed comparative analysis is then conducted against recent state-of-the-art IoV authentication and privacy-preserving schemes to demonstrate the effectiveness, scalability, and efficiency of the proposed approaches under dynamic vehicular network conditions. The results clearly highlight the superiority of the proposed frameworks in achieving secure, efficient, and privacy-aware authentication for Internet of Vehicles environments.

5.2 Analysis Of Existing Models

5.2.1 Comparative Performance Analysis of the Authentication Framework

Tables 5.1 and 5.2 present a consolidated comparative evaluation of the proposed blockchain-based privacy-preserving authentication framework against existing optimization-driven models and conventional classification techniques. The comparison is conducted using standard performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, and Matthews Correlation Coefficient (MCC), to comprehensively assess detection reliability and robustness. From the results, it is evident that the proposed OBWO-HANet consistently outperforms all benchmark optimization algorithms such as GWO-HANet, EOO-HANet, LOA-HANet, and BWO-HANet, achieving the highest accuracy of 93.8% and superior sensitivity of 93.76%, which indicates improved attack detection capability.

Similarly, when compared with traditional classifiers including CNN, MLP, Ridge classifier, and Ensemble models, the proposed framework demonstrates notable gains across all evaluation metrics, while maintaining lower false positive and false negative rates. These improvements confirm that the integration of blockchain with optimized hybrid learning significantly enhances authentication reliability and privacy preservation in IoV environments, validating the effectiveness of the proposed approach.

Table 5.1: Comparative analysis of the proposed blockchain-based privacy-preserving authentication scheme for IoV with conventional optimization algorithms

Terms	GWO-HANet	EOO-HANet	LOA-HANet	BWO-HANet	OBWO-HANet
Accuracy	88.52	86.52	91.96	88.96	93.8
Sensitivity	88.40809845	86.74077015	91.94124653	88.80508138	93.767368
Specificity	88.63361548	86.29584845	91.97904071	89.11729141	93.8331318
Precision	88.76046234	86.53465347	92.08747515	89.23015556	93.91650099
FPR	11.36638452	13.70415155	8.020959291	10.88270859	6.166868198
FNR	11.59190155	13.25922985	8.058753474	11.19491862	6.232631997
NPV	88.27780008	86.50505051	91.83098592	88.68832732	93.68209256
FDR	11.23953766	13.46534653	7.912524851	10.76984444	6.083499006
F1-Score	88.58392999	86.63758921	92.01430274	89.01711102	93.84187525
MCC	0.770399882	0.730381613	0.839193741	0.779204278	0.875995467

Table 5.2: Comparative performance analysis of the proposed blockchain-based privacy-preserving authentication scheme for IoV against conventional classification models

Terms	CNN	MLP	Ridge-classifier	Ensemble	OBWO-HANet
Accuracy	86.1	89.72	88.16	91.6	93.8

Sensitivity	85.94680429	89.75784041	88.24930528	91.62366018	93.767368
Specificity	86.25554212	89.68158001	88.06932688	91.57597743	93.8331318
Precision	86.39265762	89.8291617	88.24930528	91.69646404	93.91650099
FPR	13.74445788	10.31841999	11.93067312	8.424022572	6.166868198
FNR	14.05319571	10.24215959	11.75069472	8.376339817	6.232631997
NPV	85.80593424	89.60934354	88.06932688	91.50221506	93.68209256
FDR	13.60734238	10.1708383	11.75069472	8.303535956	6.083499006
F1-Score	86.16915423	89.79348689	88.24930528	91.66004766	93.84187525
MCC	0.722004691	0.794389628	0.763186322	0.831991584	0.875995467

5.3 Performance Evaluation of the Proposed CD-RRNN-Based Attack Detection Model

Table 5.3 presents the overall performance evaluation of the proposed Cascaded Dilated Residual Recurrent Neural Network (CD-RRNN) for malicious attack detection in Internet of Vehicles environments, in comparison with widely used learning-based classifiers such as DNN, SVM, LSTM, and conventional RNN models. The evaluation is conducted using multiple standard performance metrics to comprehensively assess detection accuracy, robustness, and reliability. As observed from the results, the proposed CD-RRNN achieves the highest accuracy of 94.34% and sensitivity of 94.32%, while maintaining significantly lower false positive and false negative rates compared to all benchmark models. These improvements indicate a strong capability to correctly identify malicious activities while minimizing misclassification of legitimate vehicular communications. The superior performance of CD-RRNN is primarily attributed to its cascaded architecture combined with dilated residual connections, which enhance long-term dependency learning and effectively capture subtle temporal attack patterns. Overall, the results confirm that the proposed attack detection mechanism provides a reliable and efficient solution for strengthening authentication and security in dynamic IoV networks.

Table 5.3: Comparative analysis of proposed system with traditional models

Terms	DNN	SVM	LSTM	RNN	CD-RRNN(Proposed Model)
Accuracy	86.8	89.16	87.04	91.74	94.34
Sensitivity	86.82016673	89.20206431	87.13775308	91.66335848	94.3231441
Specificity	86.77952439	89.11729141	86.97497	91.8178154	94.35711407
Precision	86.9582505	89.27294398	87.13775308	91.91878981	94.43561208
FPR	13.22047561	10.88270859	13.0592503	8.182184603	5.642885933
FNR	13.17983327	10.79793569	12.86224692	8.336641524	5.676855895
FOR	13.36016097	10.95449054	13.0592503	8.440514469	5.7568438
PT	0.195111666	0.174642947	0.193564606	0.14938492	0.12229579
FM	86.88918118	89.23749711	87.13775308	91.79098529	94.37936134
PLHR	6.567098587	8.196678576	6.672492759	11.20279765	16.71540861
TS	76.81770285	80.56651129	77.20717552	84.82733284	89.35690109

5.4 Summary

The proposed blockchain-based authentication and privacy-preserving frameworks demonstrated superior performance compared to conventional IoV security approaches. The OBWO-HANet–based authentication model achieved higher accuracy, sensitivity, and reliability through optimized learning and blockchain integration, while significantly reducing false detection rates. The CD-RRNN–based security framework combined with MMRFO-optimized El-Gamal encryption (OKEE) further strengthened system protection by enabling efficient attack detection, faster encryption and decryption, lower memory consumption, and enhanced robustness. Overall, the experimental results confirm that the integrated learning, optimization, and cryptographic mechanisms effectively improve authentication, intrusion resilience, and data privacy in dynamic IoV environments, outperforming existing advanced models solutions.